

# Data Engineering Workflow Platform

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Open Source Data Engineering Platform สำหรับสร้างและจัดการ Data Pipeline ตั้งแต่ Data Source → Data Lake → Transformation → Data Quality → Data Warehouse → Analytics → AI

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## 1. Project Overview

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**Data Engineering Workflow Platform (DEWP)** เป็นแพลตฟอร์มสำหรับสร้าง Data Engineering Workflow แบบครบวงจร โดยเน้นให้สามารถ

- Ingest ข้อมูล
- Transform ข้อมูล
- Validate ข้อมูล
- Orchestrate Pipeline
- Store ข้อมูล
- Monitor Pipeline
- Track Data Lineage
- ส่งข้อมูลต่อให้ BI, ML และ AI

แนวคิดหลัก:

```
DATA
↓
PIPELINE
↓
QUALITY
↓
WAREHOUSE
↓
ANALYTICS
↓
AI
```

---

## 2. Quick Start

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### 2.1 Requirements

ติดตั้งเครื่องมือพื้นฐาน:

```
brew install git python uv  
brew install --cask docker
```

ตรวจสอบ:

```
git --version  
python3 --version  
uv --version  
docker --version  
docker compose version
```

---

## 3. Create Project

---

สร้าง Project:

```
mkdir data-engineering-workflow  
cd data-engineering-workflow
```

สร้าง Python Environment:

```
uv venv --python 3.13  
source .venv/bin/activate
```

ติดตั้ง Python Packages:

```
uv pip install \  
    pandas \  
    polars \  
    sqlalchemy \  
    psycpg2-binary \  
    requests \  
    python-dotenv \  
    pydantic \  
    dbt-postgres
```

---

## 4. Project Structure

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## สร้างโครงสร้างเริ่มต้น:

```
mkdir -p \  
dags \  
ingestion \  
transformation \  
quality \  
warehouse \  
dbt/models/staging \  
dbt/models/intermediate \  
dbt/models/marts \  
data/bronze \  
data/silver \  
data/gold \  
tests \  
scripts \  
notebooks \  
docs \  
monitoring
```

## โครงสร้าง:

```
data-engineering-workflow/  
|  
├── dags/  
├── ingestion/  
├── transformation/  
├── quality/  
├── warehouse/  
├── dbt/  
│   ├── models/  
│   │   ├── staging/  
│   │   ├── intermediate/  
│   │   └── marts/  
├── data/  
│   ├── bronze/  
│   ├── silver/  
│   └── gold/  
├── tests/  
├── scripts/  
├── notebooks/  
├── docs/  
├── monitoring/  
├── docker-compose.yml  
└── requirements.txt
```

```
├── .env
├── .gitignore
└── README.md
```

---

## 5. Start Infrastructure

---

Development Infrastructure ใช้:

```
Docker
├── PostgreSQL
├── MinIO
└── Airflow
```

เริ่มระบบ:

```
docker compose up -d
```

ตรวจสอบ:

```
docker compose ps
```

หยุดระบบ:

```
docker compose down
```

---

## 6. First Pipeline

---

Pipeline ตัวแรกใช้ Sales Dataset:

```
CSV
↓
Extract
↓
Validate
↓
Bronze
```

```
↓  
Transform  
↓  
Silver  
↓  
Data Quality  
↓  
Gold  
↓  
PostgreSQL
```

---

## 7. Create Sample Data

---

สร้างไฟล์:

```
data/bronze/sales.csv
```

ข้อมูลตัวอย่าง:

```
transaction_id,transaction_date,customer_id,product_id,quantity,unit_price  
TX001,2026-09-01,C001,P001,2,100.00  
TX002,2026-09-01,C002,P002,1,250.00  
TX003,2026-09-02,C001,P003,3,150.00  
TX004,2026-09-02,C003,P001,5,100.00  
TX005,2026-09-03,C004,P002,2,250.00  
TX006,2026-09-03,C002,P003,1,150.00
```

---

## 8. Run Transformation

---

ตัวอย่าง Python:

```
python transformation/sales.py
```

ผลลัพธ์:

```
data/  
|  
└─ bronze/
```

```
└─ sales.csv
└─ silver/
   └─ sales_clean.csv
└─ gold/
   └─ sales_daily.csv
```

---

## 9. Verify the Result

---

ตรวจสอบข้อมูล:

```
cat data/silver/sales_clean.csv
```

ตรวจสอบ Gold:

```
cat data/gold/sales_daily.csv
```

ตัวอย่างผลลัพธ์:

```
transaction_date,transaction_count,units_sold,revenue
2026-09-01,2,3,450
2026-09-02,2,8,750
2026-09-03,2,3,650
```

---

## 10. Verify PostgreSQL

---

เข้า PostgreSQL:

```
docker exec -it de-postgres \
psql -U dataeng -d datawarehouse
```

ตรวจสอบ Schema:

```
\dn
```

ตรวจสอบ Tables:

```
\dt raw.*
\dt staging.*
\dt mart.*
```

Query:

```
SELECT *
FROM mart.sales_daily;
```

---

## 11. Architecture

---

ระบบประกอบด้วย:

```
graph TD
    DS[Data Sources] --> I[Ingestion]
    I --> DL[Data Lake]
    DL --> B[Bronze]
    B --> T[Transformation]
    T --> S[Silver]
    S --> DQ[Data Quality]
    DQ --> G[Gold]
    G --> DW[Data Warehouse]
    DW --> BA[BI / Analytics / ML / AI]
```

flowchart TB

```
SOURCES["DATA SOURCES"]
INGEST["INGESTION"]
LAKE["DATA LAKE MinIO"]
BRONZE["BRONZE Raw"]
TRANSFORM["TRANSFORMATION Python / SQL / dbt / Spark"]
SILVER["SILVER Cleaned"]
```

```
QUALITY["DATA QUALITY"]
GOLD["GOLD Business Ready"]
WAREHOUSE["DATA WAREHOUSE PostgreSQL"]
CONSUMPTION["BI / Analytics / ML / AI"]

SOURCES --> INGEST
INGEST --> LAKE
LAKE --> BRONZE
BRONZE --> TRANSFORM
TRANSFORM --> SILVER
SILVER --> QUALITY
QUALITY --> GOLD
GOLD --> WAREHOUSE
WAREHOUSE --> CONSUMPTION
```

---

## 12. Data Workflow

---

flowchart TD

```
START(["START"])
EXTRACT["Extract"]
VALIDATE["Validate"]
BRONZE["Load Bronze"]
CLEAN["Clean"]
TRANSFORM["Transform"]
SILVER["Load Silver"]
QUALITY["Data Quality"]
GATE{"PASS?"}
QUARANTINE["Quarantine"]
GOLD["Build Gold"]
WAREHOUSE["Load Warehouse"]
PUBLISH["Publish"]
END(["SUCCESS"])
```

```
START --> EXTRACT
EXTRACT --> VALIDATE
VALIDATE --> BRONZE
BRONZE --> CLEAN
CLEAN --> TRANSFORM
TRANSFORM --> SILVER
SILVER --> QUALITY
QUALITY --> GATE
```

```
GATE -->|YES| GOLD
GATE -->|NO| QUARANTINE
```

```
GOLD --> WAREHOUSE
```



```
WAREHOUSE --> PUBLISH
PUBLISH --> END
```

---

## 13. Bronze / Silver / Gold

---

Project ใช้ **Medallion Architecture**

flowchart LR

```
SOURCE["SOURCE"]
BRONZE["BRONZE Raw Data"]
SILVER["SILVER Cleaned Validated"]
GOLD["GOLD Business Ready"]
```

```
SOURCE --> BRONZE
BRONZE --> SILVER
SILVER --> GOLD
```

### Bronze

ข้อมูลต้นฉบับ:

```
Raw
Original
Immutable
Traceable
```

### Silver

ข้อมูลที่ผ่านการ:

```
Cleaning
Standardization
Deduplication
Validation
```

### Gold

ข้อมูลพร้อมสำหรับ:

Analytics  
Reporting  
BI  
ML  
AI

---

## 14. Data Quality

---

ตัวอย่าง Quality Rules:

```
transaction_id
  → UNIQUE

customer_id
  → NOT NULL

quantity
  → > 0

unit_price
  → >= 0

transaction_date
  → VALID DATE
```

Architecture:

flowchart TD

```
DATA["Dataset"]
SCHEMA["Schema"]
NULL["NULL"]
DUP["Duplicate"]
TYPE["Data Type"]
RANGE["Range"]
FRESH["Freshness"]
BUSINESS["Business Rule"]
GATE{"Quality Gate"}
PASS["PASS"]
FAIL["FAIL"]
QUARANTINE["Quarantine"]
```

```
DATA --> SCHEMA
SCHEMA --> NULL
NULL --> DUP
```

```
DUP --> TYPE
TYPE --> RANGE
RANGE --> FRESH
FRESH --> BUSINESS
BUSINESS --> GATE

GATE --> |PASS| PASS
GATE --> |FAIL| FAIL

FAIL --> QUARANTINE
```

---

## 15. Apache Airflow

---

Airflow ทำหน้าที่เป็น Workflow Orchestrator

flowchart TD

```
DAG["Airflow DAG"]
INGEST["Ingestion"]
BRONZE["Bronze"]
SILVER["Silver"]
DBT["dbt"]
QUALITY["Quality"]
GOLD["Gold"]
WAREHOUSE["Warehouse"]
MONITOR["Monitoring"]

DAG --> INGEST
INGEST --> BRONZE
BRONZE --> SILVER
SILVER --> DBT
DBT --> QUALITY
QUALITY --> GOLD
GOLD --> WAREHOUSE
WAREHOUSE --> MONITOR
```

Airflow รับผิดชอบ:

- Scheduling
- Task Dependency
- Retry
- Failure Handling
- Monitoring
- Alerting

## 16. dbt

---

dbt ใช้สำหรับ SQL Transformation

```
Raw
↓
Staging
↓
Intermediate
↓
Mart
```

โครงสร้าง:

```
dbt/
├── models/
│   ├── staging/
│   │   └── stg_sales.sql
│   ├── intermediate/
│   │   └── int_sales.sql
│   └── marts/
│       └── fct_sales.sql
├── tests/
└── dbt_project.yml
```

คำสั่ง:

```
dbt debug
```

```
dbt run
```

```
dbt test
```

---

## 17. Data Lake

---

## ใช้ MinIO เป็น Object Storage

```
MinIO
├── bronze
├── silver
└── gold
```

Data Flow:

flowchart LR

```
SOURCE["Data Source"]
BRONZE["MinIO Bronze"]
SILVER["MinIO Silver"]
GOLD["MinIO Gold"]
```

```
SOURCE --> BRONZE
BRONZE --> SILVER
SILVER --> GOLD
```

---

## 18. Data Warehouse

---

PostgreSQL:

```
PostgreSQL
├── raw
├── staging
├── intermediate
└── mart
```

ตัวอย่าง:

```
CREATE SCHEMA IF NOT EXISTS raw;

CREATE SCHEMA IF NOT EXISTS staging;

CREATE SCHEMA IF NOT EXISTS mart;
```

## 19. Data Lineage

---

flowchart LR

```
SOURCE["Source"]
BRONZE["Bronze"]
SILVER["Silver"]
GOLD["Gold"]
WAREHOUSE["Warehouse"]
DASHBOARD["Dashboard"]
AI["AI / RAG"]
```

```
SOURCE --> BRONZE
BRONZE --> SILVER
SILVER --> GOLD
GOLD --> WAREHOUSE
WAREHOUSE --> DASHBOARD
WAREHOUSE --> AI
```

สามารถติดตาม:

```
Source
↓
Transformation
↓
Dataset
↓
Warehouse
↓
Dashboard / AI
```

---

## 20. Monitoring

---

Monitoring ควรเก็บ:

```
Pipeline Status
Task Duration
Records Processed
Records Failed
Data Freshness
Quality Score
Error Rate
Retry Count
```

flowchart LR

```
PIPELINE["Pipeline"]
LOGS["Logs"]
METRICS["Metrics"]
ALERT["Alerts"]
DASHBOARD["Monitoring Dashboard"]
```

```
PIPELINE --> LOGS
PIPELINE --> METRICS
```

```
LOGS --> DASHBOARD
METRICS --> DASHBOARD
```

```
METRICS --> ALERT
```

## 21. Technology Stack

| Layer                  | Technology                     |
|------------------------|--------------------------------|
| Language               | Python                         |
| Query                  | SQL                            |
| Container              | Docker                         |
| Orchestration          | Apache Airflow                 |
| Transformation         | dbt                            |
| Data Lake              | MinIO                          |
| Warehouse              | PostgreSQL                     |
| Processing             | Pandas / Polars                |
| Distributed Processing | Apache Spark                   |
| Quality                | dbt Tests / Great Expectations |
| Lineage                | OpenLineage                    |
| Catalog                | DataHub                        |
| Monitoring             | Prometheus / Grafana           |
| Version Control        | Git                            |
| CI/CD                  | GitHub Actions                 |

## 22. Project Structure

---

```
data-engineering-workflow/  
├── dags/  
├── ingestion/  
├── transformation/  
├── quality/  
├── warehouse/  
  
├── dbt/  
│   ├── models/  
│   │   ├── staging/  
│   │   ├── intermediate/  
│   │   └── marts/  
│   ├── tests/  
│   └── dbt_project.yml  
  
├── data/  
│   ├── bronze/  
│   ├── silver/  
│   └── gold/  
  
├── monitoring/  
├── tests/  
├── notebooks/  
├── scripts/  
├── docs/  
  
├── docker/  
├── .github/  
│   └── workflows/  
  
├── docker-compose.yml  
├── requirements.txt  
├── .env.example  
├── .gitignore  
├── README.md  
└── LICENSE
```

---

## 23. Data Model

---

ตัวอย่าง Sales Data Warehouse:

erDiagram



```

CUSTOMER {
    string customer_id PK
    string customer_name
    string segment
}

PRODUCT {
    string product_id PK
    string product_name
    string category
    decimal price
}

DATE {
    int date_key PK
    date date
    int year
    int month
    int day
}

SALES {
    string transaction_id PK
    string customer_id FK
    string product_id FK
    int date_key FK
    int quantity
    decimal unit_price
    decimal total_amount
}

CUSTOMER ||--o{ SALES : buys
PRODUCT ||--o{ SALES : contains
DATE ||--o{ SALES : occurs

```

## 24. AI / RAG Integration

Data Engineering Platform สามารถเป็นฐานข้อมูลสำหรับ AI:

flowchart TB

```

DATA["Enterprise Data"]
PLATFORM["Data Engineering Platform"]
LAKE["Data Lake"]
WAREHOUSE["Data Warehouse"]
KNOWLEDGE["Knowledge Layer"]
RAG["RAG"]
AGENT["AI Agent"]

```

DATA --> PLATFORM

PLATFORM --> LAKE

PLATFORM --> WAREHOUSE

LAKE --> KNOWLEDGE

WAREHOUSE --> KNOWLEDGE

KNOWLEDGE --> RAG

RAG --> AGENT

รองรับ:

- Knowledge Base
- RAG
- Embedding Pipeline
- Vector Search
- AI Agent
- Document Processing
- Retrieval Evaluation

---

## 25. Distributed Processing

---

เมื่อข้อมูลมีขนาดใหญ่ สามารถเพิ่ม Apache Spark:

flowchart LR

```
SOURCE["Large Data"]
LAKE["Data Lake"]
SPARK["Apache Spark"]
SILVER["Silver"]
GOLD["Gold"]
WAREHOUSE["Warehouse"]
```

```
SOURCE --> LAKE
LAKE --> SPARK
SPARK --> SILVER
SILVER --> GOLD
GOLD --> WAREHOUSE
```

แนวทางเลือก Processing Engine:

Small Data

↓

Python / Pandas

Medium Data

↓

Polars / SQL

Large Data

↓

Apache Spark

---

## 26. Development Roadmap

---

Phase 01

Foundation

↓

Phase 02

Data Ingestion

↓

Phase 03

Data Lake

↓

Phase 04

ETL / ELT

↓

Phase 05

Airflow

↓

Phase 06

dbt

↓

Phase 07

Data Quality

↓

Phase 08

Data Warehouse

↓

Phase 09

Data Lineage

↓

Phase 10

Data Catalog

↓

Phase 11

Monitoring

↓

Phase 12

CI/CD

↓

Phase 13  
Spark  
↓  
Phase 14  
Production  
↓  
Phase 15  
AI / RAG / Agent

---

## 27. Production Architecture

---

flowchart TB

USERS["Users / Applications"]

subgraph SOURCES["DATA SOURCES"]

DB["Databases"]

API["APIs"]

FILE["Files"]

EVENT["Events"]

end

subgraph PLATFORM["DATA ENGINEERING PLATFORM"]

INGEST["Ingestion"]

LAKE["Data Lake"]

BRONZE["Bronze"]

SILVER["Silver"]

QUALITY["Data Quality"]

GOLD["Gold"]

WH["Data Warehouse"]

LINEAGE["Lineage"]

CATALOG["Catalog"]

MONITOR["Monitoring"]

end

AIRFLOW["Apache Airflow"]

subgraph CONSUMPTION["CONSUMPTION"]

BI["BI"]

ANALYTICS["Analytics"]

ML["ML"]

AI["AI / RAG / Agent"]

end

DB --> INGEST

API --> INGEST

FILE --> INGEST

EVENT --> INGEST

INGEST --> LAKE  
LAKE --> BRONZE  
BRONZE --> SILVER  
SILVER --> QUALITY  
QUALITY --> GOLD  
GOLD --> WH

WH --> BI  
WH --> ANALYTICS  
WH --> ML  
WH --> AI

AIRFLOW -. -> INGEST  
AIRFLOW -. -> BRONZE  
AIRFLOW -. -> SILVER  
AIRFLOW -. -> GOLD  
AIRFLOW -. -> QUALITY  
AIRFLOW -. -> WH

QUALITY --> LINEAGE  
WH --> LINEAGE  
LINEAGE --> CATALOG

AIRFLOW -. -> MONITOR  
QUALITY -. -> MONITOR  
WH -. -> MONITOR

USERS --> CONSUMPTION

---

## 28. Production Checklist

---

### Infrastructure

- ☐ Docker
- ☐ PostgreSQL
- ☐ MinIO
- ☐ Airflow

### Pipeline

- ☐ Ingestion
- ☐ Bronze
- ☐ Silver
- ☐ Gold
- ☐ Warehouse

### Quality

- ☐ Schema Validation
- ☐ NULL Check
- ☐ Duplicate Check
- ☐ Business Rules
- ☐ Freshness

## Operations

- ☐ Scheduling
- ☐ Retry
- ☐ Logging
- ☐ Monitoring
- ☐ Alerting

## Governance

- ☐ Data Lineage
- ☐ Data Catalog
- ☐ Metadata
- ☐ Data Ownership

## Engineering

- ☐ Git
- ☐ Tests
- ☐ CI/CD
- ☐ Docker
- ☐ Documentation

## Advanced

- ☐ Spark
- ☐ Distributed Processing
- ☐ ML Pipeline
- ☐ RAG
- ☐ AI Agent

---

# 29. Final Vision

---

flowchart TB

```
DATA["DATA"]  
INGEST["INGEST"]  
STORE["STORE"]  
PROCESS["PROCESS"]
```

```
VALIDATE["VALIDATE"]
ORCHESTRATE["ORCHESTRATE"]
GOVERN["GOVERN"]
MONITOR["MONITOR"]
ANALYTICS["ANALYTICS"]
AI["AI"]
```

```
DATA --> INGEST
INGEST --> STORE
STORE --> PROCESS
PROCESS --> VALIDATE
VALIDATE --> ORCHESTRATE
ORCHESTRATE --> GOVERN
GOVERN --> MONITOR
```

```
MONITOR --> ANALYTICS
ANALYTICS --> AI
```

### Vision:

**Build → Automate → Validate → Observe → Govern → Scale → AI**

Data Engineering Workflow Platform ถูกออกแบบให้เริ่มจาก Pipeline ขนาดเล็กบนเครื่อง Local ได้ทันที และสามารถขยายไปสู่ Data Platform, Distributed Processing และ AI Data Infrastructure ได้ในอนาคต.

## 30. License

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